

MGMS IM GF USHAKOVA, Russian Federation

WMO: 200870

Lat: 79.551N

Lon: 90.572E

Elev: 8

StdP: 101.23

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-38.3	-36.3	-40.6	0.1	-38.1	-38.6	0.1	-36.2	15.1	-20.9	13.8	-21.8	3.8	140	1.180

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.7	4.1	3.1	3.0	2.3	2.2	1.7	3.2	3.9	2.5	2.9	1.8	2.1	5.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2.5	4.5	3.3	1.9	4.3	2.6	1.4	4.2	1.9	15.2	4.0	13.9	3.0	12.8	2.2	12.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	12.7	11.4	DB	-39.0	6.4	3.1	2.1	-41.3	8.0	-43.1	9.2	-44.9	10.4	-47.2	12.0	(4)
(5)			WB	-39.0	5.1	3.0	2.0	-41.2	6.5	-43.0	7.7	-44.7	8.8	-46.9	10.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-12.6	-25.1	-25.6	-24.5	-18.3	-9.2	-1.2	0.7	0.3	-2.0	-8.0	-16.8	-22.3
	DBStd	11.36	7.01	7.02	6.51	5.72	4.36	2.01	1.37	1.34	2.57	5.54	6.33	6.78
	HDD10.0	8249	1088	996	1070	850	596	337	289	300	359	558	805	1001
	HDD18.3	11291	1347	1229	1328	1100	854	587	547	558	609	816	1055	1259
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.7	5.9	5.7	5.8	5.6	5.0	5.1	5.0	5.3	6.1	6.9	6.2	6.3
Precipitation	PrecAvg	175	10	9	9	8	9	16	28	29	22	14	10	11
	PrecMax	386	52	23	26	50	30	55	148	123	46	23	19	29
	PrecMin	85	1	2	1	0	2	2	1	2	3	3	2	2
	PrecStd	60	10	5	6	10	7	12	28	21	13	6	5	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.4	-3.5	-3.1	-0.8	0.5	3.2	6.8	4.8	2.8	0.9	-0.9	-2.6
		MCWB	-2.7	-3.9	-3.5	-1.3	0.0	1.9	5.1	3.8	2.5	0.2	-1.4	-3.0
	2%	DB	-6.3	-7.5	-7.0	-4.9	-0.5	2.2	4.8	3.6	2.1	0.4	-1.9	-5.1
		MCWB	-6.6	-7.9	-7.3	-5.4	-1.0	1.2	3.4	2.9	1.9	-0.1	-2.3	-5.5
	5%	DB	-10.7	-11.4	-10.8	-8.0	-1.7	1.6	3.6	2.7	1.4	-0.4	-3.7	-8.4
		MCWB	-10.9	-11.6	-11.1	-8.3	-2.3	0.8	2.7	2.2	1.2	-0.8	-4.1	-8.7
	10%	DB	-15.4	-15.4	-15.0	-10.5	-2.9	1.0	2.5	2.1	0.8	-1.5	-7.0	-12.8
		MCWB	-15.7	-15.7	-15.3	-11.0	-3.5	0.5	1.9	1.7	0.6	-2.0	-7.4	-13.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.7	-3.8	-3.5	-1.4	0.2	2.2	5.3	4.1	2.7	0.6	-1.2	-2.8
		MCDB	-2.5	-3.4	-3.2	-1.0	0.4	2.8	6.6	4.5	2.7	0.7	-1.0	-2.7
	2%	WB	-6.6	-7.8	-7.2	-5.3	-0.9	1.4	3.6	3.1	2.0	0.0	-2.3	-5.4
		MCDB	-6.4	-7.5	-7.0	-5.0	-0.6	1.9	4.5	3.5	2.1	0.3	-1.8	-5.1
	5%	WB	-10.9	-11.7	-11.0	-8.3	-2.2	1.0	2.7	2.3	1.3	-0.8	-4.1	-8.7
		MCDB	-10.6	-11.4	-10.7	-8.0	-1.7	1.4	3.5	2.6	1.4	-0.5	-3.7	-8.4
	10%	WB	-15.6	-15.6	-15.3	-10.9	-3.5	0.6	2.0	1.7	0.7	-1.9	-7.4	-13.1
		MCDB	-15.4	-15.4	-15.0	-10.5	-3.0	0.9	2.4	2.0	0.8	-1.5	-6.9	-12.8
Mean Daily Temperature Range	5% DB	MDBR	5.9	5.6	5.9	5.4	3.9	2.0	1.7	1.7	1.9	3.3	4.9	5.4
		MCDBR	9.3	9.6	9.4	6.7	3.6	2.3	3.6	2.6	1.6	2.1	4.8	9.2
		MCWBR	9.3	9.5	9.5	6.7	3.5	1.9	2.8	2.2	1.6	2.3	4.9	9.1
	5% WB	MCDBR	9.3	9.6	9.5	6.7	3.6	2.1	3.3	2.5	1.6	2.2	4.9	9.3
		MCWBR	9.3	9.5	9.5	6.7	3.6	1.8	2.7	2.2	1.6	2.4	4.9	9.1
Clear-Sky Solar Irradiance	taub	N/A	N/A	0.245	0.289	0.314	0.307	0.327	0.314	0.257	N/A	N/A	N/A	N/A
	taud	N/A	N/A	1.934	2.008	2.058	2.192	2.289	2.407	2.273	N/A	N/A	N/A	N/A
	Ebn at Noon	0	0	485	717	783	821	780	722	540	0	0	0	0
	Edh at Noon	0	0	50	93	111	102	89	65	38	0	0	0	0
All-Sky Solar Radiation	RadAvg	0.00	0.00	0.81	3.47	5.77	6.43	4.93	2.54	0.81	0.00	0.00	0.00	0.00
	RadStd	0.00	0.00	0.08	0.20	0.40	0.52	0.62	0.29	0.12	0.00	0.00	0.00	0.00

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+1.67	+2.48	+1.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page